



QUANTUM MACHINE LEARNING

MODULE 1: INTRODUCTION

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- 3 Motivation: Why Quantum Machine Learning?
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COURSE OBJECTIVES

- CO1 Understand the mathematical foundations of quantum computing
- CO2 Explain core quantum algorithms relevant to Machine learning
- CO3 Compare classical and quantum ML approaches
- CO4 Evaluate effectiveness of quantum ML approaches

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TEXT AND REFERENCE BOOKS

TEXT BOOKS

- T1 Maria Schuld, & Francesco Petruccione. "Machine Learning with Quantum Computers", Springer, 2021.
- T2 Thomas G Wong "Introduction to Classical and Quantum Computing" 2022

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EVALUATION SCHEDULE

No	Name	Mode	Weight
EC1	Quiz I and II	Online	10 %
	Lab Assignment I	Online	10 %
	Lab Assignment II	Online	10 %
EC2	Mid-sem	Closed book	30%
EC3	Comprehensive	Open book	40%

EVALUATION COMPUTATION

EC1 computation

$EC1 = \text{Best (Quiz I, Quiz II)} + \text{Assignment I} + \text{Assignment II}$

NO makeups for Quizzes and Assignments

Total

$\text{Total} = EC1 + EC2 + EC3$

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QUANTUM COMPUTING: HYPE, REALITY, AND CURIOSITY

What We Often Hear

- Quantum computers are **much faster** than classical machines
- They can **break all encryption**
- Qubits are "0 and 1 at the same time"
- They will soon replace laptops
- You need a PhD in Physics to understand them

Let's Ask

- What is actually true?
- What is exaggerated?
- Where does machine learning fit in?



Quantum Mechanics

Study of matter and light at atomic and subatomic scales. Foundation of quantum information science.

Quantum Computation

Information processing using quantum mechanical systems (Nielsen & Chuang).

Quantum Machine Learning

Explores how quantum computation can enhance learning from data.



Quantum technologies being explored:

- Quantum sensing systems (incl. measuring instruments)
- Quantum information systems (incl. cryptography)
- Quantum communication (incl. networks)
- Quantum computers
- Quantum-inspired systems

Different models of quantum computing:

- Quantum simulators (incl. HPC accelerators)
- Quantum adiabatic / annealing computers
- Topological quantum computers
- Quantum circuit / gate models
- Universal quantum computers (Quantum Turing Machines)



Major National / Regional Programs

- USA** — National Quantum Initiative (multi-agency: DOE, NSF, NIST)
- China** — State-led national programs (computing, communication, sensing)
- European Union** — Quantum Flagship (1B) + national missions
- United Kingdom** — National Quantum Technologies Programme
- Canada** — Quantum Supercluster + federal R&D
- Australia, Japan, South Korea** — National quantum roadmaps



Government of India — NQM (2023–2031)

- Approved budget: 6003.65 crore (multi-year mission)
- Focus areas:
 - Quantum computing (50 - 1000 qubit targets)
 - Quantum communication (secure networks)
 - Quantum sensing & metrology
 - Quantum materials & devices
- Establishment of Thematic Hubs across institutions
- Goal: Build indigenous quantum ecosystem (hardware + software)

India joins the group of countries with a dedicated national quantum mission.

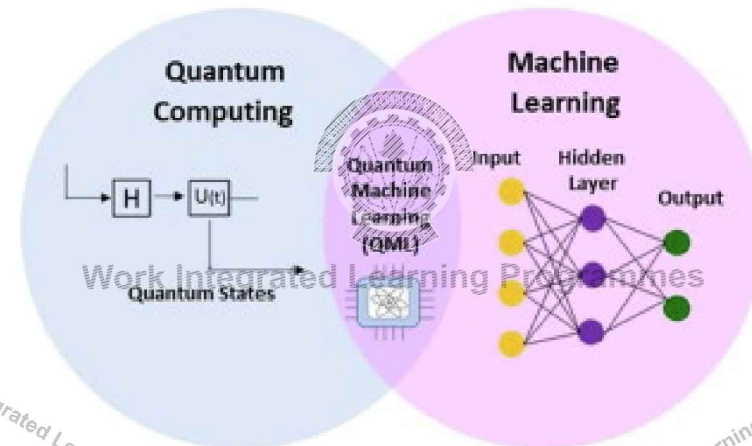


Hardware (Gate-Based / Annealing)

- IBM, Google, Rigetti (Superconducting qubits)
- IonQ, Honeywell (Trapped ions)
- Microsoft (Topological / Majorana approach)
- D-Wave (Quantum annealing)

Software & Hybrid Platforms

- Qiskit (IBM)
- Cirq (Google)
- PennyLane (Xanadu)
- TensorFlow Quantum
- Azure Quantum



- **Finance** — risk modeling, credit scoring, fraud detection, portfolio ML
- **Cybersecurity** — anomaly detection, intrusion detection, malware classification
- **Healthcare** — disease classification (images/signals), triage models, outcome prediction
- **Drug discovery & chemistry** — learning molecular properties, generative molecule design
- **Materials / manufacturing** — defect detection (vision), predictive maintenance, process optimization
- **Transport / logistics** — demand forecasting, routing prediction, supply-chain anomaly detection



Potential Benefits

- Natural handling of high-dimensional vector spaces
- Efficient sampling and probabilistic modeling
- New algorithmic paradigms beyond classical heuristics
- Integration with physical modeling and simulation

Practical Risks / Challenges

- Hardware noise and scalability limits (NISQ constraints)
- High development and infrastructure cost
- Limited skilled workforce and tooling maturity
- Uncertain real-world speedups for many ML tasks

BRIEF TIMELINE

- **1981–1994** Foundations of quantum computing (Feynman, Deutsch, Shor, Grover)
- **2014** Formal quantum SVM proposal (Rebentrost et al.)
- **2016** Biamonte et al. - Quantum ML as a field
- **2018** Quantum kernel methods (Havlíček et al., Nature)
- **2019** Quantum “supremacy” experiments (Google)
- **2020–2022** Rapid growth of Variational Quantum Circuits (VQCs) Hybrid quantum–classical learning
- **2023–2024** Focus shifts to NISQ-aware models and error mitigation
- **Present** QML research emphasizes:
 - ▶ Trainability
 - ▶ Expressivity vs classical models
 - ▶ Realistic hardware constraints

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WHERE CLASSICAL COMPUTATION GETS HARD

In many ML pipelines, the main costs are:

- **High-dimensional representations** (explicit or implicit feature maps)
- **Similarity computations** across large datasets (e.g., kernels / nearest neighbors)
- **Sampling/optimization** in complex probabilistic models

Key question

Can quantum systems represent and manipulate certain high-dimensional structures more naturally?

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QUANTUM COMPUTERS

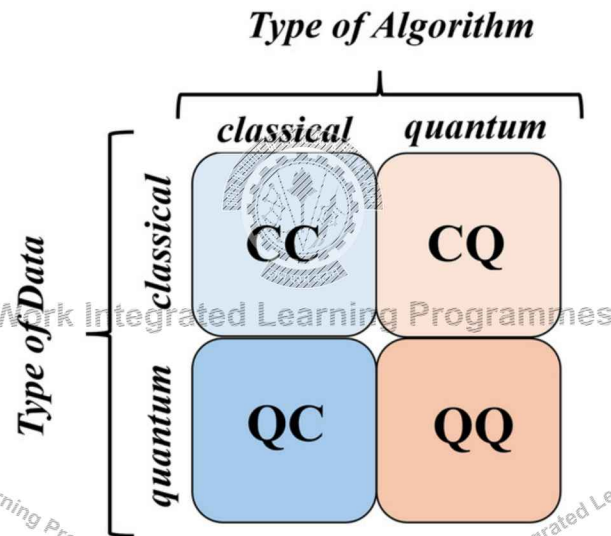


Generality

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FOUR APPROACHS TO QML



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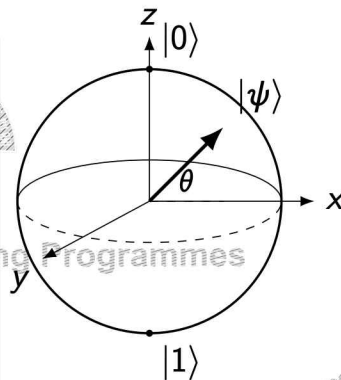
BLOCH SPHERE INTUITION

Parameterization

Any pure qubit state can be written as:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

- θ : polar angle
- ϕ : azimuthal angle



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QUANTUM BUILDING BLOCK



Qubit?

A qubit is like a classical bit — but it can be in a blend of 0 and 1 at the same time.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

- α, β are numbers that determine the state
- $|\alpha|^2$ = probability of measuring 0
- $|\beta|^2$ = probability of measuring 1
- The total probability must equal 1

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TOY EXAMPLE SETUP

A tiny supervised dataset (Mini Titanic)

Each input is a 2D feature vector:

$$\mathbf{x}_m = (x_{m0}, x_{m1})^T,$$

with label $y_m \in \{0, 1\}$ indicating survival (1) or not (0).

Features

- Feature 0: ticket price (scaled to $[0, 1]$)
- Feature 1: cabin number (scaled to $[0, 1]$)



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TOY EXAMPLE DATASET

Mini-dataset for the quantum classifier example

(lr)2-3(lr)4-6	Raw data		Preprocessed data		
	Price	Cabin	Price	Cabin	Survival
Passenger 1	8,500	0910	0.85	0.36	1 (yes)
Passenger 2	1,200	2105	0.12	0.84	0 (no)
Passenger 3	7,800	1121	0.78	0.45	?

Task

Given Passenger 3 with $\mathbf{x} = (0.78, 0.45)$, predict Survival using the labelled points.



DISTANCE BASED CLASSICAL MODELS

Nearest neighbor (NN)

Assign the label of the closest training point (or majority among k nearest).

Weighted variant used in the toy example

Instead of hard voting, each training point influences the decision with a weight that decreases with distance.

Why weighted?

It creates a smooth, probability-like output - and maps neatly to quantum overlap computations.



SQUARED-DISTANCE CLASSIFIER

Given a new input x and training points x_m , define a distance-based weight:

$$\gamma_m = 1 - \frac{1}{c} \|x - x_m\|^2,$$

where c is a constant (chosen so weights stay in a useful range).

Interpretation

- If x is close to x_m , then $\|x - x_m\|^2$ is small $\Rightarrow \gamma_m$ is large.
- If x is far, then γ_m becomes smaller.

Connection ahead

This is closely related to **kernel ideas**.



WORKED EXAMPLE

Let the query be Passenger 3:

$$x = (0.78, 0.45), \quad x_1 = (0.85, 0.36), \quad x_2 = (0.12, 0.84).$$

Compute the needed inner products

$$\langle x, x \rangle = 0.78^2 + 0.45^2 = 0.8109$$

$$\langle x_1, x_1 \rangle = 0.85^2 + 0.36^2 = 0.8521$$

$$\langle x_2, x_2 \rangle = 0.12^2 + 0.84^2 = 0.7200$$

$$\langle x, x_1 \rangle = 0.78 \cdot 0.85 + 0.45 \cdot 0.36 = 0.8250$$

$$\langle x, x_2 \rangle = 0.78 \cdot 0.12 + 0.45 \cdot 0.84 = 0.4716$$



DISTANCE DEPENDS ON INNER PRODUCTS

For real vectors,

$$\|x - x_m\|^2 = (x - x_m)^T (x - x_m) = x^T x + x_m^T x_m - 2x^T x_m.$$

So the classifier needs quantities like $x^T x_m = \langle x, x_m \rangle$.

Why this matters for QML

Quantum mechanics naturally computes and exposes **overlaps** $\langle x | x_m \rangle$ via interference/measurement.

Takeaway

The toy example is intentionally chosen because it “fits the quantum formalism.”



WORKED EXAMPLE

Compute squared distances

$$\|x - x_1\|^2 = 0.8109 + 0.8521 - 2(0.8250) = 0.0130$$

$$\|x - x_2\|^2 = 0.8109 + 0.7200 - 2(0.4716) = 0.5877$$

Convert distance to similarity

$$\gamma_m = 1 - \frac{1}{c} \|x - x_m\|^2$$

Choose $c \geq \max_m \|x - x_m\|^2$ so that $0 \leq \gamma_m \leq 1$.



FROM WEIGHTS TO PREDICTION (PROBABILITY-LIKE OUTPUT)

For binary labels $y_m \in \{0, 1\}$, the prediction combines all training samples using their weights:

$$\hat{y}(x) = \frac{\sum_{m=1}^M \gamma_m y_m}{\sum_{m=1}^M \gamma_m}.$$

Interpretation

- Distances determine weights γ_m for each training point.
- Closer points receive larger weights.
- The classifier combines information from all labelled samples.



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PREDICTION PROPERTIES

Properties

- If nearby points have label 1, $\hat{y}(x)$ increases.
- If nearby points have label 0, $\hat{y}(x)$ decreases.
- When labels conflict, $\hat{y}(x)$ lies between 0 and 1.

So, $\hat{y}(x)$ behaves like a probability-like output indicating the likelihood of survival.

Quantum mapping

The classifier depends on quantities related to $\|x - x_m\|^2$, which can be written using inner products.

Quantum algorithms can help compute such overlaps efficiently.



ALGEBRA TO QUANTUM COMPUTATION

We now ask:

- How can a quantum circuit compute something like an overlap?
- How can quantum states be combined to reveal similarities?

Key idea

Quantum circuits manipulate amplitudes so that useful outcomes become more likely when measured.

CIRCUIT WARM-UP: CREATING SUPERPOSITION



Result

Applying a Hadamard gate creates an equal combination of two states:

$$0 \xrightarrow{H} \frac{0+1}{\sqrt{2}}$$

(the symbols 0 and 1 represent the two possible states).

Point to remember

The Hadamard gate splits the computation into two simultaneous possibilities.

HADAMARD GATE

Computational basis states

- A single qubit has two basis states: $|0\rangle$ and $|1\rangle$
- Vector representation:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hadamard transformation

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The gate creates an equal superposition of the basis states.

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

INTERFERENCE INTUITION: ADDITIONS AND CANCELLATIONS

Suppose two amplitudes a and b combine during a quantum computation. After interference:

some outcomes $\propto (a + b)$, others $\propto (a - b)$.

Why this matters for ML

Many ML decisions involve comparing similarity:

- "Is x closer to x_1 or x_2 ?"
- Which overlap with the data is larger?

Quantum interference can make measurement probabilities depend on these additions and cancellations.

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QUANTUM SQUARED-DISTANCE CLASSIFIER

Lets see how Hadamard interference can compute the prediction of the squared-distance classifier using a quantum circuit.

- 1 **Step A:** Additional preprocessing (normalize vector length to 1)
- 2 **Step B:** Encode data into a quantum state (amplitude encoding)
- 3 **Step C:** Apply Hadamard to induce interference between amplitude blocks
- 4 **Step D:** Measure/post-select to extract the prediction

STEP A: UNIT-NORM PROJECTION

Normalize each input vector so its length equals 1:

$$\|x\| = 1, \|x_m\| = 1.$$

Geometrically, all points lie on a unit circle (in 2D). After normalization, similarity depends mainly on **angles between vectors**.

Caveat

Normalization can discard magnitude information; later chapters discuss workarounds.

NUMERICAL: UNIT-NORM DATA

After Step A, each passenger feature vector is normalised to length 1.

$$x^{(1)} = (0.921, 0.390), x^{(2)} = (0.141, 0.990), x^{(3)} = (0.866, 0.500)$$

Check the norms

$$\|x^{(1)}\|^2 = 0.921^2 + 0.390^2 \approx 1$$

$$\|x^{(2)}\|^2 = 0.141^2 + 0.990^2 \approx 1, \|x^{(3)}\|^2 = 0.866^2 + 0.500^2 \approx 1$$

After normalization, similarity is determined mainly by the **angle between vectors**.

GEOMETRIC VIEW: UNIT-NORM PROJECTION

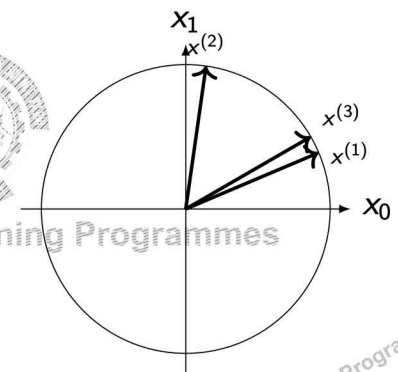
After Step A, all feature vectors lie approximately on the unit circle:

$$x^{(1)} = (0.921, 0.390)$$

$$x^{(2)} = (0.141, 0.990)$$

$$x^{(3)} = (0.866, 0.500)$$

- Magnitude information is suppressed
- Angle becomes the key similarity cue





ANGLE-BASED SIMILARITY

If $\|x\| = \|y\| = 1$, then

$$x^T y = \cos(\theta)$$

where θ is the angle between the vectors.

Interpretation

- Large overlap $\Rightarrow$ small angle $\Rightarrow$ high similarity
- After normalization, squared distance becomes

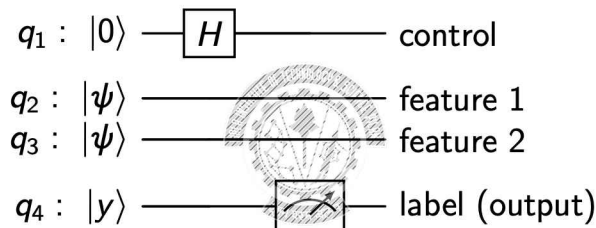
$$\|x - y\|^2 = 2 - 2 \cos(\theta)$$

- Angles matter in both: classical cosine similarity and quantum state overlaps.



QUBIT ROLES IN THE CIRCUIT

How data and labels are represented



Interpretation

- q_2, q_3 encode the feature vector (amplitudes)
- q_1 creates superposition (parallel paths)
- q_4 stores the class label (0 or 1)
- Measurement is performed only on q_4



STEP B: DATA ENCODING

Amplitude encoding

Encode a normalized feature vector into amplitudes of a quantum state:

$$x = (x_0, x_1, \dots, x_{d-1}) \mapsto |x\rangle = \sum_{i=0}^{d-1} x_i |i\rangle$$

Example ($d = 4$)

Two qubits represent four amplitudes:

$$|x\rangle = x_0 |00\rangle + x_1 |01\rangle + x_2 |10\rangle + x_3 |11\rangle$$

Key point

Only $\log_2(d)$ qubits are needed to represent d amplitudes. Preparing such states efficiently can be difficult.



BLOCK PAIRING FOR INTERFERENCE

Amplitude structure: We repeat the query point twice

$$(x^{(1)}, x^{(2)}, x, x)$$

Paired during Hadamard

$$(x^{(1)}, x), (x^{(2)}, x)$$

After Hadamard

$$(x^{(1)} + x), (x^{(1)} - x)$$

$$(x^{(2)} + x), (x^{(2)} - x)$$

Each training point is compared with the input simultaneously.

NUMERICAL: AMPLITUDE VECTOR



Step B encodes the data into amplitudes. Using the unit-normalized vectors from Step A, the amplitude vector becomes

$$\alpha = \frac{1}{\sqrt{4}} (0.921, 0.390, 0.141, 0.990, 0.866, 0.500, 0.866, 0.500)^T$$

Interpretation

- First 4 entries: training data features
- Last 4 entries: copies of the new input features
- Factor $1/\sqrt{4}$: keeps the full amplitude vector normalised

The circuit will now act on these amplitudes directly.

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STEP C: INTERFERENCE

Block interference

Arrange amplitudes into two blocks.

Applying a Hadamard gate creates combinations:

$$\text{block}_0 \rightarrow (\text{block}_0 + \text{block}_1)$$

$$\text{block}_1 \rightarrow (\text{block}_0 - \text{block}_1)$$

What Hadamard enables This creates sums and differences of amplitude patterns, allowing the circuit to compare similarities.

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REPRESENTATION VS PREPARATION



Representation With $n = \log_2 d$ qubits, we can represent a d -dimensional normalized vector in amplitudes.

Preparation Preparing an arbitrary amplitude state can require $\mathcal{O}(d)$ time.

If state preparation dominates the runtime, any theoretical speedup may disappear.

Takeaway

Quantum advantage claims must consider state preparation and measurement sampling costs.

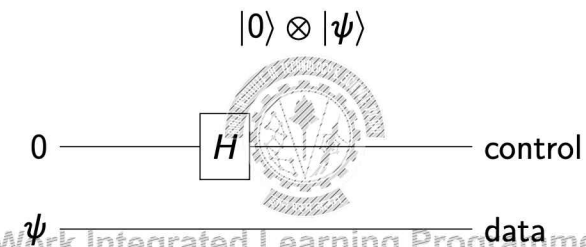
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STEP C: CIRCUIT SKETCH

Initial state



State after Hadamard

$$\frac{1}{\sqrt{2}} (|0\rangle|\psi\rangle + |1\rangle|\psi\rangle)$$

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NUMERICAL: HADAMARD ON BLOCKS

Applying Hadamard to the data

$$\frac{1}{\sqrt{8}}(0.921 + 0.866), \quad \frac{1}{\sqrt{8}}(0.390 + 0.500)$$

$$\frac{1}{\sqrt{8}}(0.141 + 0.866), \quad \frac{1}{\sqrt{8}}(0.990 + 0.500)$$

and in the other branch,

$$\frac{1}{\sqrt{8}}(0.921 - 0.866), \quad \frac{1}{\sqrt{8}}(0.390 - 0.500), \dots$$

STEP D: MEASUREMENT INTUITION

After interference, the system is in a superposed state

We measure the **control qubit**.

- Measurement collapses the quantum state to either:

$$|0\rangle|\phi_0\rangle \quad \text{or} \quad |1\rangle|\phi_1\rangle$$

- The probability of each outcome depends on amplitudes:

$$P(0) = \text{depends on overlap / interference}$$

- These probabilities encode similarity between states

Measurement converts quantum amplitudes into classical probabilities.

STEP D: POST-SELECTION

Measure the first qubit. If the outcome is $|0\rangle$, keep the run; otherwise repeat.

What is being selected?

After interference:

$$|0\rangle(\text{sum branch}) + |1\rangle(\text{difference branch})$$

- $|0\rangle$ branch contains **sums** ($a + b$) $\rightarrow$ encodes similarity
- $|1\rangle$ branch contains **differences** ($a - b$) $\rightarrow$ not used for prediction

Why both branches exist?

- Quantum operations must be **unitary (reversible)**
- So both $(a + b)$ and $(a - b)$ are created

Post-selection introduces a repeat-until-success cost.

REVERSIBLE VS IRREVERSIBLE COMPUTATION

Classical computation (irreversible)

- Example: $(a, b) \rightarrow a + b$
- Output loses information about inputs
- Cannot recover a and b uniquely

Quantum computation (must be reversible)

- Operations must preserve information
- Example: $(a, b) \rightarrow (a + b, a - b)$
- Both outputs together allow recovery of inputs

Quantum gates are reversible transformations (called **unitary**), so they cannot discard information.

NUMERICAL: OUTPUT PROBABILITIES

After post-selection, measure the last qubit.

For label 0:

$$p(q_4 = 0) = \frac{1}{8\chi} (|0.141 + 0.866|^2 + |0.990 + 0.500|^2) \approx 0.448$$

For label 1:

$$p(q_4 = 1) = \frac{1}{8\chi} (|0.921 + 0.866|^2 + |0.390 + 0.500|^2) \approx 0.552$$

The quantum circuit predicts label 1 with higher probability.

SUMMARY

Algorithm structure

- Normalize inputs
- Encode into quantum states
- Create interference (Hadamard)
- Measure → obtain probabilities

FROM MEASUREMENT TO PROBABILITY

After interference, we measure the control qubit

- Measurement produces outcomes: 0 or 1
- Each outcome occurs with some probability
- These probabilities depend on interference between states

$$P(0), P(1)$$

Key Idea

Quantum computation produces **probabilities**, not direct labels.

WHERE DOES THE PREDICTION COME FROM?

After interference and post-selection, measurement probabilities can be interpreted as

$$Pr(\text{survival}) \approx \hat{y}(x) \in [0, 1]$$

Why probability is natural

Quantum systems produce probabilities when measured. This aligns naturally with ML models that produce **soft predictions** rather than only hard labels.

Student takeaway

In QML, the model output is often a probability extracted from measurement statistics.

NUMERICAL: MATCH TO THE CLASSICAL CLASSIFIER

Final result from the quantum circuit

$$p(q_4 = 0) \approx 0.448, \quad p(q_4 = 1) \approx 0.552$$

Meaning

- Label 1 ("survival") has the higher probability
- So Passenger 3 is classified closer to Passenger 1 than Passenger 2
- This matches the classical squared-distance classifier

Big takeaway

The toy quantum circuit reproduces the squared-distance classifier using **encoding + interference + measurement**.

PRACTICE PROBLEM

Given

Training data:

$$x^{(1)} = (3, 4), \quad y^{(1)} = 1$$

$$x^{(2)} = (0, 5), \quad y^{(2)} = 0$$

New input:

$$x = (4, 3)$$

Tasks

- Step A: Normalize all vectors
- Step B: Construct amplitude vector
- Step C: Apply Hadamard (block sum/difference)
- Step D: Compute probabilities and predict label

Goal Predict the label of the new input using the quantum classifier.

SOLUTION: STEP A & B

Step A: Normalization

$$\|x^{(1)}\| \Rightarrow (0.6, 0.8)$$

$$\|x^{(2)}\| \Rightarrow (0, 1)$$

$$\|x\| \Rightarrow (0.8, 0.6)$$

Step B: Amplitude Encoding

$$\alpha = \frac{1}{\sqrt{4}}(0.6, 0.8, 0, 1, 0.8, 0.6, 0.8, 0.6)^T$$

- First 4 entries: training data
- Last 4 entries: input (duplicated)

SOLUTION: STEP C (INTERFERENCE)

Hadamard mixes amplitude blocks:

$$(a, b) \rightarrow \frac{1}{\sqrt{2}}(a + b, a - b)$$

Sum branch (used after post-selection)

$$\frac{1}{\sqrt{8}}(0.6 + 0.8) = \frac{1.4}{\sqrt{8}} \quad \frac{1}{\sqrt{8}}(0.8 + 0.6) = \frac{1.4}{\sqrt{8}}$$

$$\frac{1}{\sqrt{8}}(0 + 0.8) = \frac{0.8}{\sqrt{8}} \quad \frac{1}{\sqrt{8}}(1 + 0.6) = \frac{1.6}{\sqrt{8}}$$

SOLUTION: STEP D (MEASUREMENT)

Compute probabilities

Label 1:

$$\left(\frac{1.4}{\sqrt{8}}\right)^2 + \left(\frac{1.4}{\sqrt{8}}\right)^2 = 0.49$$

Label 0:

$$\left(\frac{0.8}{\sqrt{8}}\right)^2 + \left(\frac{1.6}{\sqrt{8}}\right)^2 = 0.40$$

Normalize:

$$p(1) = 0.551, \quad p(0) = 0.449$$

Final Prediction

$$y = 1$$

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INSIGHTS

- We chose squared distance because it aligns with quantum overlaps.
- Interference (Hadamard) is powerful but hard to engineer meaningfully.
- Normalization changes what information is retained (angles vs magnitudes).
- Data encoding/state preparation is a central challenge (later revisited).
- This toy logic repeats across QML: encode $\rightarrow$ interfere $\rightarrow$ measure.

Takeaway

The toy example is **not** about beating classical ML today; it is about learning the QML design vocabulary.

COMMON MISCONCEPTIONS

- 1 “Quantum = faster always” $\rightarrow$ Not true. Advantage is conditional.
- 2 “Encoding is free” $\rightarrow$ Often the hardest step.
- 3 “One run gives exact answer” $\rightarrow$ Measurements are probabilistic; need many shots.
- 4 “Toy example = practical algorithm” $\rightarrow$ It is a teaching scaffold.

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- 1 Course Logistics
- 2 Introduction
- 3 Motivation: Why Quantum Machine Learning?
- 4 Toy Example Setup
- 5 Classical ML: Squared Distance classifier
- 6 Bridge to Quantum
- 7 Quantum Processing on Toy Example
- 8 Insights and Misconceptions
- 9 Summary and Self Practice

RECAP

Squared-distance classifier

$$\gamma_m = 1 - \frac{1}{c} \|x - x_m\|^2, \quad \hat{y}(x) = \frac{\sum_m \gamma_m y_m}{\sum_m \gamma_m}.$$

Distance expands into inner products:

$$\|x - x_m\|^2 = x^T x + x_m^T x_m - 2x^T x_m.$$

Quantum idea

Amplitude encoding + Hadamard interference can transform amplitude blocks so that post-selected measurement statistics approximate the classifier output.



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Thank you :)



QUANTUM MACHINE LEARNING

MODULE 2: ML TO QML

BITS Pilani

Prof. Neha Vinayak



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- 1 Orientation
- 2 ML as a Mathematical Problem
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MODULE 2 PLAN

- Quick recap of ML concepts from Chapter 2: data, model, loss, risk, generalisation
- Reframe ML in a mathematical form suitable for quantum models
- Understand kernels and neural networks as modelling paradigms
- Identify where quantum computing fits in the ML pipeline

Design Philosophy

ML is assumed known. This module builds the conceptual bridge to QML.



WHY QML NEEDS ML

- QML still follows the ML paradigm: data $\rightarrow$ model $\rightarrow$ loss $\rightarrow$ optimisation
- Only specific parts of the pipeline change with quantum computing
- Understanding ML abstractions makes QML intuitive

Key Message

Quantum computing does not replace ML — it introduces new representations and similarity computations.



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REFRAMING MACHINE LEARNING



Instead of focusing on algorithms (SVM, CNN, etc.), we use a unified abstraction:



- x : input data
- θ : model parameters
- f : parameterised function

Key Idea

All ML models can be viewed as function approximators.

THE LEARNING PROBLEM



Given a dataset:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$$

Goal:

$$f(x, \theta) \approx y$$

- Learn from finite samples
- Generalise to unseen data

ML AS OPTIMISATION



Training is solving:

$$\theta^* = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n \ell(f(x_i, \theta), y_i)$$

- ℓ : loss function
- Search over parameter space

Interpretation

Learning = selecting the best function from a model family.

QUANTUM PREVIEW



- Classical model: $f(x, \theta)$
- Quantum model: $U(\theta)$ (parameterised circuit)
- Output obtained via measurement

Important Insight

The learning objective remains the same — only the model representation changes.



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Work Integrated Learning Programmes

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DATA

Dataset:

$$\mathcal{D} = \{(x_i, y_i)\}$$

- Finite samples approximate real-world processes
- Quality and quantity of data affect learning

QML Perspective

Classical data must be encoded into quantum states before processing.

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THREE INGREDIENTS OF ML

Every ML method can be described using:

- Data: input-output pairs
- Model family: set of candidate functions
- Loss: measure of prediction error

Quantum Connection

QML changes the model family and representation of data, but keeps the loss classical.

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MODEL FAMILY

A model family is a set of functions:

$$\{f_{\theta} : \theta \in \Theta\}$$

- Defines hypothesis space
- Examples: linear models, kernels, neural networks

QML Perspective

Model = quantum circuit with tunable parameters.

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LOSS FUNCTION

Loss compares prediction with ground truth:

$$\ell(f_{\theta}(x), y)$$

- Regression: squared error
- Classification: cross-entropy

Important

Loss function remains classical even in QML.



EMPIRICAL RISK

Training objective:

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i)$$

- Computed on training data
- Used as proxy for true risk

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GENERALISATION

- Goal: perform well on unseen data
- Training success alone is insufficient

Generalisation gap = test loss – train loss

Insight

Low training error does not guarantee good generalisation.

OVERFITTING

- Model becomes too specialised to training data
- Test error increases with complexity

Trade-off

Balance between underfitting and overfitting.

QUANTUM CONNECTION

- Quantum measurements introduce statistical noise
- Finite shots affect output estimates
- Device noise impacts training stability

Implication

Generalisation in QML is affected by both data and hardware noise.

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MOTIVATION: WHY FEATURE MAPS?

Many real-world datasets are not linearly separable in input space.

- Example: fraud detection, medical diagnosis, speech signals
- Simple models fail in raw feature space

Idea:

$$x \mapsto \phi(x)$$

- Map data into a higher-dimensional space
- In that space, simple models (e.g., linear) may work well

Insight

Complexity can be shifted from the model to the representation.

KERNEL TRICK: LEARNING VIA SIMILARITY

Instead of explicitly computing $\phi(x)$, we define:

$$k(x, x') = \langle \phi(x), \phi(x') \rangle$$

- Measures similarity between inputs
- Avoids explicit high-dimensional computation

Key idea:

- Learning depends only on pairwise similarities
- No need to construct $\phi(x)$ explicitly

Real-world analogy

Recommendation systems: users are matched based on similarity, not explicit features.

KERNEL MATRIX AND LEARNING

For training points $x_1, \dots, x_n$:

$$K_{ij} = k(x_i, x_j)$$

- Encodes all pairwise relationships
- Many algorithms (e.g., SVM) depend only on K

Interpretation:

- Each row = similarity profile of a data point
- Learning = exploiting structure in similarity matrix

Insight

Kernel methods transform ML into a problem of similarity computation.

QUANTUM FEATURE MAPS

Classical:

$$x \rightarrow \phi(x)$$

Quantum:

$$x \rightarrow |\psi(x)\rangle$$

- Encoding circuit maps classical data to quantum state
- State lives in a high-dimensional Hilbert space

Key Insight

Quantum systems naturally provide extremely high-dimensional feature spaces.

QUANTUM KERNELS

Quantum kernel:

$$k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$$

- Estimated using quantum circuits + measurements
- No explicit feature construction needed

Why important?

- Potentially hard-to-compute similarities classically
- One of the earliest practical QML approaches

Takeaway

Kernel methods provide a natural entry point into QML.



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REPRESENTATION LEARNING

Neural networks learn hierarchical features:

- Early layers: simple patterns (edges, frequencies)
- Middle layers: shapes, motifs
- Final layers: decision boundaries

Key Insight

Good representations simplify the learning problem.

QML analogy:

- Quantum circuits transform input states into useful representations

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NEURAL NETWORKS AS PARAMETRIC MODELS

Neural networks represent functions:

$$f(x, \theta)$$

- Built from layers of linear + nonlinear transformations
- Highly expressive function families

Key property:

- Can approximate very complex mappings

Example

Image recognition: raw pixels $\rightarrow$ object labels

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VARIATIONAL QUANTUM CIRCUITS (VQCs)

VQCs are parameterised quantum models:

- Input encoded into quantum state
- Circuit applies parameterised gates
- Output obtained via measurement

$$\text{Model} = U(\theta)$$

Key Idea

Quantum circuits act as learnable functions similar to neural networks.

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HYBRID QUANTUM-CLASSICAL MODELS

Typical workflow:

- Classical preprocessing (feature extraction)
- Quantum circuit (representation + transformation)
- Classical post-processing (decision layer)

Reality

Near-term QML is hybrid due to hardware constraints.

Example: CNN features → quantum circuit → classifier

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LEARNING AS AN OPTIMISATION LOOP

All ML models follow:

- Forward pass → prediction
- Loss computation → error
- Parameter update → improvement

Core Idea

Learning is iterative refinement of parameters.

GRADIENT-BASED OPTIMISATION

Parameter update:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \hat{R}(\theta)$$

- η : learning rate
- Gradient gives direction of steepest descent

Challenge

Optimisation landscape may be complex and non-convex.



OPTIMISATION IN QML

Key differences in quantum setting:

- Outputs estimated via finite measurements
- Gradients computed using sampling-based methods
- Noise affects optimisation stability

Implication

Optimisation in QML is inherently stochastic and noisy.



CLASSICAL VS QUANTUM PIPELINES

Classical ML:

- Input → Features → Model → Prediction → Loss

Quantum ML:

- Input → Encoding → Circuit → Measurement → Loss

Key Difference

Quantum ML introduces encoding and measurement stages.



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WHAT CHANGES IN QML?

- Representation: classical vectors → quantum states
- Computation: classical operations → quantum circuits
- Noise: deterministic → probabilistic + hardware noise

Impact

Model behaviour and training dynamics change significantly.



WHAT REMAINS THE SAME?

- Loss minimisation
- Generalisation objectives
- Training-validation-test paradigm

Key Insight

QML is still fundamentally ML — the learning logic is unchanged.



PROBABILISTIC NATURE OF OUTPUTS

Quantum models produce probabilistic outputs:

- Measurement yields samples
- Expectation values estimated from repeated runs

Interpretation

Outputs are estimates, not exact values.

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SHOTS AND NOISE

- More shots → better estimates
- Fewer shots → faster but noisier training
- Hardware noise introduces bias

Trade-off

Accuracy vs computation time vs noise robustness

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CLASSICAL VS QUANTUM MAPPING

Classical ML	Quantum ML
$f(x, \theta)$	$U(\theta)$
features	quantum state
inner product	state overlap
output	measurement
training	parameter tuning

Mental Model

Replace classical function with quantum circuit, rest remains similar.

KEY TAKEAWAYS

- ML = function approximation + optimisation
- Kernels: similarity-based learning
- Neural networks: representation learning
- QML modifies representation and model family
- Near-term QML is hybrid and measurement-driven

POINTS TO PONDER

Points to Ponder

- If ML is just $f(x, \theta)$, what limits classical models — data, computation, or representation?
- Kernel methods rely on similarity: can quantum systems compute richer similarities than classical methods?
- Neural networks learn representations implicitly — can quantum circuits learn fundamentally different representations?
- How do noise and probabilistic outputs affect learning in QML compared to classical ML?

Next Session: Mathematical Foundations for QML

Thank you :)



QUANTUM MACHINE LEARNING

MODULE 3: MATHEMATICAL FOUNDATIONS FOR QML

Prof. Neha Vinayak

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1 Introduction

2 Complex Vector Spaces

3 Bra-Ket Notation and Hilbert Space

4 Tensor Product

5 Unitary Transformations

WHY THIS MODULE MATTERS

Context in the course

- Quantum computing is not "mysterious mathematics."
- It is mainly linear algebra with three important changes:
 - vectors can be **complex-valued**,
 - valid states must satisfy a **normalization** constraint,
 - allowed transformations must be **unitary**.

NEW VS FAMILIAR

What looks new

- Symbols like $|\psi\rangle$, $\langle\psi|$, $\otimes$, and $U^\dagger$
- Complex amplitudes instead of only real numbers
- Probability coming from squared magnitudes rather than direct values

What is actually familiar

- Vectors still represent states or data
- Matrices still represent transformations
- Inner products still represent similarity / overlap
- Norm still measures size

So this is not a separate mathematics subject; it is a constrained version of linear algebra used for computation.

BRIDGE ML TO QML

Classical ML language	Quantum ML language
feature vector $\mathbf{x} \in \mathbb{R}^n$	state vector $ \psi\rangle \in \mathbb{C}^n$
dot product $\mathbf{x}^T \mathbf{y}$	inner product $\langle \mathbf{x} \mathbf{y} \rangle$
matrix transform $A\mathbf{x}$	unitary transform $U \psi\rangle$
normalized embedding	normalized quantum state
similarity / kernel	overlap / quantum kernel
network output score	probability / expectation

ROADMAP OF THIS MODULE

We will cover five building blocks

- 1 Complex vector spaces
- 2 Hilbert spaces
- 3 Tensor products
- 4 Bra-ket notation
- 5 Unitary transformations

And we will keep asking two questions throughout

- What does this mean mathematically?
- Why does this matter for QML?

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COMPLEX NUMBERS: CLASSICAL VS QUANTUM REPRESENTATION

Classical ML

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad x_i \in \mathbb{R}$$

Quantum setting



$$|\psi\rangle = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \quad \alpha_i \in \mathbb{C}$$

Why complex numbers appear

- Magnitude determines probability ($|\alpha|^2$)
- Phase controls interference between amplitudes
- Both are required to describe quantum systems
- So a quantum state is still a vector, but in a richer space than classical ML.

WHY REAL NUMBERS ARE NOT ENOUGH

Classical representation

- Real numbers encode only magnitude (value on a line)

Quantum systems behave like waves

- Need to represent:
 - Magnitude (how much)
 - Phase (where in the cycle)

Limitation of real numbers

- Can represent only two phases: 0° (positive) and 180° (negative)
- Cannot represent arbitrary phase differences

Key insight

- Quantum computation relies on interference
- Interference requires full phase control
- $\Rightarrow$ Complex numbers are necessary, not optional

REVIEW OF COMPLEX NUMBERS FOR QUANTUM COMPUTING

General form

$$z = a + ib, \quad i^2 = -1$$

Real and imaginary parts

$$\Re(z) = a, \quad \Im(z) = b$$

Example

$$z = 3 + 4i$$

Simple intuition

- Real numbers move on a line.
- Complex numbers move on a plane.
- Quantum state evolution can therefore involve rotation-like behavior in a richer space.

WHAT DOES i REPRESENT?

Geometric meaning

- Multiplication by -1 flips direction (180° rotation)
- Multiplication by i rotates by 90° in the complex plane

Cycle of rotations

$$1 \xrightarrow{\times i} i \xrightarrow{\times i} -1 \xrightarrow{\times i} -i \xrightarrow{\times i} 1$$

Interpretation

- Real axis $\rightarrow$ horizontal direction
- Imaginary axis $\rightarrow$ vertical direction
- Complex number = magnitude + direction (phase)

Key idea

- i is not abstract — it represents rotation and phase

COMPLEX CONJUGATE AND MAGNITUDE

Complex conjugate

$$z = a + ib \Rightarrow z^* = a - ib$$

Magnitude squared

$$|z|^2 = z^* z = (a - ib)(a + ib) = a^2 + b^2$$

Example

$$z = 1 + i, \quad z^* = 1 - i$$

$$|z|^2 = (1 - i)(1 + i) = 2$$

Why conjugation matters

- It ensures lengths and probabilities remain real and non-negative.
- Without conjugation, inner products in complex spaces would not behave properly.

POLAR FORM AND PHASE INTUITION

Polar form of a complex number

$$z = r e^{i\theta}$$

Meaning

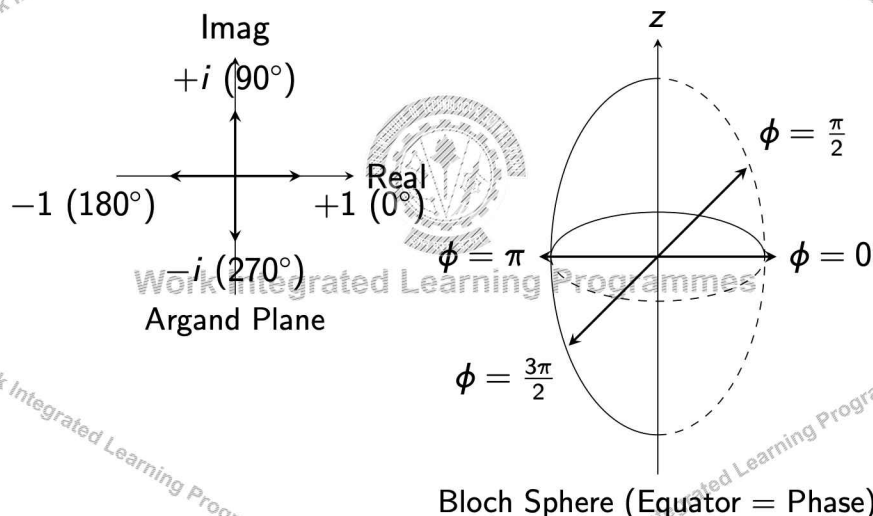
- r is the magnitude
- θ is the phase angle

Why phase is important

- Two amplitudes with the same magnitude can behave differently if their phases differ.
- When amplitudes combine, phases can reinforce or cancel each other.
- Phase enables constructive and destructive interference

Phase can be thought of as hidden geometric information carried by the state.

PHASE: ARGAND PLANE VS BLOCH SPHERE



$+i$ vs $-i \rightarrow$ opposite directions $\rightarrow$ different interference

BLOCH SPHERE INTUITION: AMPLITUDE VS PHASE

- θ controls the **amplitude split** between $|0\rangle$ and $|1\rangle$.
 - ϕ controls the **relative phase** between the two components.
 - Changing θ changes how much of the state points toward $|0\rangle$ versus $|1\rangle$.
 - Changing ϕ rotates the state **around the sphere** without changing that amplitude balance.
- Two states can have the same θ , but different ϕ , so they are **not the same quantum state**.

EXAMPLE OF A COMPLEX QUANTUM STATE

Given state

$$|\psi\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$$

Magnitude-squared

$$\left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad \left| \frac{i}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Normalization

$$\frac{1}{2} + \frac{1}{2} = 1$$

Interpretation

- Equal probability of measuring $|0\rangle$ and $|1\rangle$
- Second amplitude has phase factor i

Take Away

- Same probabilities $\neq$ same quantum state
- Phase affects future evolution (interference)

PRACTICE: CHECKING A COMPLEX STATE

Question Is the following a valid normalized state?

$$|\psi\rangle = \begin{bmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{bmatrix}$$

PRACTICE: CHECKING A COMPLEX STATE

Question Is the following a valid normalized state?

$$|\psi\rangle = \begin{bmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{bmatrix}$$

Solution

$$\left| \frac{1+i}{2} \right|^2 = \frac{1^2 + 1^2}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\left| \frac{1-i}{2} \right|^2 = \frac{1^2 + (-1)^2}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2} = 1$$

Answer

- Yes, it is a valid quantum state.

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KETS: COLUMN VECTORS



A ket is a column vector representing a quantum state

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \alpha, \beta \in \mathbb{C}$$

The standard basis kets

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Any qubit state is a linear combination of these

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Example

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

BRAS: THE CONJUGATE TRANSPOSE



A bra is the conjugate transpose ($\dagger$) of a ket

Step 1 - Transpose: turn the column into a row

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \xrightarrow{\text{transpose}} [\alpha \ \beta]$$

Step 2 - Conjugate: of every entry

$$[\alpha \ \beta] \xrightarrow{\text{conjugate}} [\alpha^* \ \beta^*] = \langle\psi|$$

In one step: $\langle\psi| = |\psi\rangle^\dagger$

Worked examples

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \langle 0| = [1 \ 0] \quad |i\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} \Rightarrow \langle i| = \frac{1}{\sqrt{2}} [1 \ -i]$$

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \langle 1| = [0 \ 1] \quad |\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \langle\psi| = [\alpha^* \ \beta^*]$$

Rule

Going from $|\psi\rangle$ to $\langle\psi|$: transpose the vector AND conjugate every entry. Going from $\langle\psi|$ to $|\psi\rangle$: the same operation applied again reverses it.

WHY THE BRA MUST BE CONJUGATED

What goes wrong without conjugation?

Take $|x\rangle = \begin{bmatrix} 1 \\ i \end{bmatrix}$. Compute its "length" naively using the ordinary transpose:

$$|x\rangle^\top |x\rangle = [1 \ i] \begin{bmatrix} 1 \\ i \end{bmatrix} = 1^2 + i^2 = 1 + (-1) = 0$$

A non-zero vector has zero length - the geometry is broken.

With the conjugate transpose (bra):

$$\langle x|x\rangle = [1 \ -i] \begin{bmatrix} 1 \\ i \end{bmatrix} = (1)(1) + (-i)(i) = 1 + 1 = 2$$

Length = $\sqrt{2}$ ✓ Non-zero, real, positive - as required.

BRA-KET: TWO PRODUCTS



There are two ways to combine a bra and a ket

Inner product (bra-ket) Outer product (ket-bra)

$\langle x|y\rangle \rightarrow \text{scalar}$ $|x\rangle\langle y| \rightarrow \text{matrix}$



- Row vector $\times$ column vector
 - Measures **overlap** between states
 - Used for: similarity, probability, norm
- Column vector $\times$ row vector
 - Constructs **operators and projectors**
 - Used for: measurement, transformations

Quick example

$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \quad (\text{scalar})$

$|0\rangle\langle 1| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (\text{matrix})$

HILBERT SPACE: THE SPACE KETS LIVE IN



Euclidean space $\mathbb{R}^n$:

- Real vectors + dot product
- Dot product gives: length, angle, perpendicularity

A **Hilbert space** is the complex generalisation:

- Complex vectors + inner product
- Inner product gives: the same tools, for complex vectors

	Euclidean space	Hilbert space
Entries	Real	Complex
Similarity tool	Dot product	Inner product
Length	$\sqrt{\mathbf{v} \cdot \mathbf{v}}$	$\sqrt{\langle \psi \psi \rangle}$
Perpendicularity	$\mathbf{u} \cdot \mathbf{v} = 0$	$\langle \phi \psi \rangle = 0$

INNER PRODUCT: DEFINITION



Definition

For two states $|\phi\rangle = [\phi_1, \phi_2, \dots, \phi_n]^T$ and $|\psi\rangle = [\psi_1, \psi_2, \dots, \psi_n]^T$:

$$\langle \phi | \psi \rangle = \phi_1^* \psi_1 + \phi_2^* \psi_2 + \dots + \phi_n^* \psi_n = \sum_{i=1}^n \phi_i^* \psi_i$$

The left state is always conjugated. This is what makes it work for complex entries.

Three key properties

- 1 **Conjugate symmetry:** $\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$
- 2 **Linearity:** $\langle \phi | \alpha \psi_1 + \beta \psi_2 \rangle = \alpha \langle \phi | \psi_1 \rangle + \beta \langle \phi | \psi_2 \rangle$
- 3 **Positivity:** $\langle \psi | \psi \rangle \geq 0$, and $= 0$ only if $|\psi\rangle = 0$

INNER PRODUCT: WORKED EXAMPLE

Given

$$|x\rangle = \begin{bmatrix} 1 \\ i \end{bmatrix}, \quad |y\rangle = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 1 — Convert $|x\rangle$ to a bra

$$\langle x| = [1 \quad -i]$$

Step 2 — Multiply

$$\langle x|y\rangle = [1 \quad -i] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (-i)(1) = 1 - i$$

Verify conjugate symmetry: compute $\langle y|x\rangle$

$$\langle y|x\rangle = [1 \quad 1] \begin{bmatrix} 1 \\ i \end{bmatrix} = 1 + i = (1 - i)^* = \langle x|y\rangle^* \quad \checkmark$$

Transition probability between the two states:

$$|\langle x|y\rangle|^2 = |1 - i|^2 = 1^2 + 1^2 = 2.$$

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INNER PRODUCT AS A SIMILARITY MEASURE

Geometric interpretation

- $|\langle \phi|\psi\rangle|^2 = 1 \Rightarrow$ states are identical
- $|\langle \phi|\psi\rangle|^2$ close to 1 $\Rightarrow$ states are very similar
- $|\langle \phi|\psi\rangle|^2$ close to 0 $\Rightarrow$ states are very different
- $\langle \phi|\psi\rangle = 0 \Rightarrow$ states are orthogonal - maximally different

Example: partial overlap

$$|x\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |y\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\langle x|y\rangle = \left[\frac{1}{\sqrt{2}} \quad \frac{1}{\sqrt{2}} \right] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \quad |\langle x|y\rangle|^2 = \frac{1}{2}$$

The states are neither identical nor orthogonal — there is partial overlap.

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ORTHOGONALITY: DEFINITION AND CHECK

Definition

Two states $|\phi\rangle$ and $|\psi\rangle$ are **orthogonal** if: $\langle \phi|\psi\rangle = 0$

Geometric meaning

- Zero overlap - states share no common component
- They carry **completely independent** information
- Measurement can **perfectly distinguish** them

Worked Example: Standard basis

$$\langle 0|1\rangle = [1 \quad 0] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \quad \checkmark$$

Worked Example: Hadamard basis

$$\langle +|- \rangle = \frac{1}{\sqrt{2}} [1 \quad 1] \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{1}{2}(1 - 1) = 0 \quad \checkmark$$

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ORTHONORMAL BASES: THE FULL CONDITION

A basis is orthonormal if every pair satisfies:

$$\langle i|j\rangle = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Every vector is **normalised** AND every pair is **orthogonal** - both at once.

Three standard orthonormal bases for a single qubit

Z-basis

$$\{|0\rangle, |1\rangle\}$$

$$\langle 0|1\rangle = 0 \quad \checkmark$$

X-basis

(Hadamard)

$$\{|+\rangle, |-\rangle\}$$

$$\langle +|-\rangle = 0 \quad \checkmark$$

Y-basis

$$\{|i\rangle, |-i\rangle\}$$

$$\langle i|-i\rangle = 0 \quad \checkmark$$

Orthonormal bases are **not unique**.

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WHY BASES MATTER

The same state measured in two different bases

State: $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

Measure in Z-basis $\{|0\rangle, |1\rangle\}$

$$P(0) = |\langle 0|+\rangle|^2 = \frac{1}{2}$$

$$P(1) = |\langle 1|+\rangle|^2 = \frac{1}{2}$$

Outcome: **completely random**

Choosing a measurement basis = choosing what features to extract from the state.

The wrong basis extracts no useful information even from a well-prepared state

Measure in X-basis

$$\{|+\rangle, |-\rangle\}$$

$$P(+) = |\langle +|+\rangle|^2 = 1$$

$$P(-) = |\langle -|+\rangle|^2 = 0$$

Outcome: **certain**

EXPANDING ANY STATE

Given an orthonormal basis $\{|e_1\rangle, \dots, |e_n\rangle\}$, any state can be written as:

$$|\psi\rangle = \sum_{i=1}^n \alpha_i |e_i\rangle, \text{ where } \alpha_i = \langle e_i|\psi\rangle$$

The amplitude for each basis state is an **inner product**

Example $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

Example $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$

$$\alpha_0 = \langle 0|\psi\rangle = \langle 0|\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle\right) = \frac{\sqrt{3}}{2} \underbrace{\langle 0|0\rangle}_1 + \frac{1}{2} \underbrace{\langle 0|1\rangle}_0 = \frac{\sqrt{3}}{2}$$

$$\alpha_1 = \langle 1|\psi\rangle = \frac{1}{2}, \quad P(0) = \frac{3}{4}, \quad P(1) = \frac{1}{4}$$

Connection to Measurement and QML

$$P(\text{outcome } i) = |\langle e_i|\psi\rangle|^2 = |\alpha_i|^2.$$

In quantum models, the circuit output is a state $|\psi(\theta)\rangle$. The prediction is read out by computing $|\langle e_i|\psi(\theta)\rangle|^2$ - inner products with measurement basis vectors.

OUTER PRODUCT AS AN OPERATOR

The ket-bra product produces a matrix

$$|x\rangle\langle y| = \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} c^* & d^* \end{bmatrix} = \begin{bmatrix} ac^* & ad^* \\ bc^* & bd^* \end{bmatrix}$$

Contrast with the inner product

- Bra-ket $\langle x|y\rangle \rightarrow$ **scalar** (a number)
- Ket-bra $|x\rangle\langle y| \rightarrow$ **matrix** (an operator acting on states)

OUTER PRODUCT AS A PROJECTION OPERATOR

Projector onto $|0\rangle$

$$|0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Apply to a general state

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow |0\rangle\langle 0|\psi\rangle = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} = \alpha|0\rangle$$

- Extracts the $|0\rangle$ component of the state
- Removes the $|1\rangle$ component entirely
- Acts as a **filter** along one basis direction

PRACTICE PROBLEMS

Q1 — Bra conversion

Write the bra $\langle\psi|$ for $|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1+i \\ 1-i \end{bmatrix}$.

Q2 — Inner product and orthogonality

Are $|\phi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}$ and $|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ 1 \end{bmatrix}$ orthogonal?

Q3 — Orthonormality check

Do $|u_1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $|u_2\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ 1 \end{bmatrix}$ form an orthonormal basis?

Q4 — State expansion

Given $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, find $\alpha_0 = \langle 0|\psi\rangle$ and $\alpha_1 = \langle 1|\psi\rangle$ using inner products and verify $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

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TENSOR PRODUCT: WHY WE NEED A NEW PRODUCT

Problem

- One qubit lives in a 2-dimensional state space.
- Two qubits do not live in another 2-dimensional space.
- We need a rule for combining systems.

That rule is the tensor product $|a\rangle \otimes |b\rangle$

- If one qubit has dimension 2, then two qubits have dimension $2 \times 2 = 4$.
- Three qubits have dimension $2^3 = 8$.
- Multi-qubit systems grow exponentially in dimension.
- This is one source of quantum representational richness.



STEP-BY-STEP RULE

Take two vectors

$$|a\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad |b\rangle = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Then

$$|a\rangle \otimes |b\rangle = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{bmatrix}$$

What is happening

- Every entry of the 1st vector multiplies the entire 2nd vector and constructs a larger joint state vector.
- This is not ordinary matrix multiplication.

EXAMPLE: 2-QUBIT HILBERT SPACE



$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle$$

$$|0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle$$

Similarly

$$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

SEPARABLE STATES AND ENTANGLED STATES

Separable state (no actual interaction yet)

$$|\psi\rangle = |a\rangle \otimes |b\rangle$$

Entangled state (is where interaction happens)

$$|\psi\rangle \neq |a\rangle \otimes |b\rangle$$

Example of an entangled state

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Tensor products create the space in which entanglement can exist.
- Not every 2-qubit state can be factorized into two independent 1-qubit states.

SEPARABLE VS ENTANGLED: HOW TO CHECK

Step 1: Write state as vector

$$|\psi\rangle = \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix}$$

Step 2: Quick test (for 2-qubit pure states only)

$$\alpha_{00}\alpha_{11} \stackrel{?}{=} \alpha_{01}\alpha_{10}$$

- Equal $\rightarrow$ separable
- Not equal $\rightarrow$ entangled

This shortcut is useful only for 2-qubit pure states. For larger systems or mixed states, separability testing is more involved.